

KARTIKEYA KUMARIA

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EDUCATION

University of California, Santa Cruz
B.S. in Computer Science

Santa Cruz, CA
Mar 2026

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

AI/ML: LLM APIs (Anthropic Claude), Retrieval-Augmented Generation (RAG), Embeddings, AI Agents, Model Context Protocol (MCP), Prompt Engineering, Vector Databases (pgvector, HNSW), LLM Evaluation

Frameworks & Libraries: React.js, Next.js, Node.js, Express, FastAPI, Django REST Framework (DRF), Sequelize

Tools & Platforms: Docker, AWS, GCP, Azure, Git, GitHub, CI/CD, CodeQL

Databases: PostgreSQL, MySQL, MongoDB, Firestore, pgvector

EXPERIENCE

Redminds

Remote

AI Enabler Apprentice

Jan 2026 – Jul 2026

- Built production case management features (autosave, advanced search, dynamic validation, schema migrations), improving data completeness across the platform.
- Implemented mobile layouts across 10+ production pages (dashboards, messaging, analytics, authentication, case management) while passing CodeQL checks.
- Extended PostgreSQL schemas and RESTful APIs, integrating Gemini API with FastAPI microservices to extract structured healthcare data from clinical PDFs.

UCSC Blueprint

Santa Cruz, CA

Software Developer

Feb 2024 – Feb 2026

- Built and launched **YFIOB**, a production full-stack K–12 career platform using React, Next.js, Firebase Authentication and Firestore.
- Mentored 5+ developers through code reviews, technical documentation and collaborative development using Git workflows.

PACR

Remote

Full Stack Development Intern

Jul 2025 – Dec 2025

- Integrated RESTful search APIs into 10+ React and Next.js components, reducing latency by 35% and increasing engagement by 25%.
- Developed 8+ production pages with persistent authentication, asynchronous data loading and responsive user workflows.

SKYIT

Remote

Backend Developer Intern

Apr 2024 – Aug 2024

- Resolved production API ingestion failures across Azure-hosted MySQL/CSV pipelines, validating multi-tenant DRF schemas to restore onboarding within 2 days.
- Designed and tested 30+ of the backend team's 401 total DRF API endpoints (auth, CSV ingestion) using DRF serializers and `rest_framework_csv`.

CodeDay

Remote

Software Engineer Intern

Oct 2023 – Dec 2023

- Implemented a Node.js Mocha test case (kWh→MJ→BTU conversion, 76-row CSV fixture) for Open Energy Dashboard; merged via Pull Request #1087, documented in a write-up.
- Collaborated with a 3-person open-source contributor team across 10 commits on a feature branch, from initial implementation through peer review to merge.

PROJECTS

AI Nexus | *Next.js, TypeScript, Express, FastAPI, Anthropic Claude, PostgreSQL, pgvector, MCP, Docker*

May 2026 – Present

- Built a full-stack AI engineering platform spanning 10 interactive chapters that demonstrate RAG, embeddings-based retrieval, AI agents, MCP and Anthropic Claude API integration.
- Architected a polyglot microservices system across 4 containerized services (Next.js, Express, FastAPI, MCP server) exposing 15 REST endpoints, deployed via Docker Compose with guides for AWS, Azure and GCP.
- Implemented 7 from-scratch algorithms for cost optimization and quality assurance, including BPE tokenization, semantic caching, TextRank summarization and ROUGE-L/embedding-based evaluation, citing 40+ academic and industry sources.